DKIM Bulk Signature Verifier

User Guide & Options Reference

Version 1.7.2 | Build 2026-11-05 | verify_dkim_bulk.py

Bulk DKIM verification of .eml evidence files

Every DKIM signature verified - RSA and Ed25519 (RFC 8463)
Three public-key sourcing modes (live DNS or preserved local key files)
Signature expiry, identity and key-flag evaluation
Every public key used preserved, with a key manifest

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Software: Apache License 2.0 | Documentation: CC BY 4.0 | Provided as-is, without warranty

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1. Overview

DKIM Bulk Signature Verifier is a Python tool that verifies the DomainKeys Identified Mail (DKIM) signatures of every `.eml` file in a folder. For each signature it recomputes the body hash, rebuilds the signed header block, checks the cryptographic signature against the signing domain's public key, and evaluates the signature's identity and time tags and the key's own flags. Results are written to CSV ledgers, supported by per-message logs, optional extracted artifacts and a copy of every public key used.

A valid DKIM signature shows that the signed headers and body have not been altered since the message was signed by the domain in the d= tag. In an investigation this helps establish whether an email produced as evidence is an unaltered copy of what the signing server sent.

What the tool does for each .eml file

- Calculates MD5 and SHA-256 hashes of the original file (evidence integrity).
- Normalises LF-only or mixed line endings to CRLF on a working copy; the original file is never changed.
- Optionally scans the message body against regex keyword patterns and records any matches.
- Finds **every** `DKIM-Signature` header (Section 5) and, for each one:
  - locates the public key according to the selected key mode, and saves a copy to keys/;
  - canonicalises the body (simple or relaxed), computes the body hash and compares it with bh=;
  - rebuilds the signed header block from the h= list and verifies the b= signature;
  - evaluates i=, t= and x= (Section 6) and the key record's flags (Section 4.5).
- Reports PASS if at least one signature passes, with per-signature details, expiry for every signature, and a log.

Supported algorithms and key formats

Item	Supported
Signature algorithms	rsa-sha256; rsa-sha1 (deprecated, flagged in Notes); ed25519-sha256 (RFC 8463)
Canonicalisation	simple/simple, relaxed/relaxed and mixed combinations (c= tag); defaults to simple/simple
Body length limit	l= tag honoured; the number of unsigned trailing bytes is reported in Notes
Signature tags evaluated	i= (identity), t= (signing time), x= (expiry)
Key record tags evaluated	k= (type), t=y (testing), t=s (strict identity), h= (hashes), s= (service), n= (notes)
Key formats	Raw DNS TXT record (v=DKIM1; k=...; p=...), zone-file style line, bare base64, PEM (BEGIN PUBLIC KEY / BEGIN RSA PUBLIC KEY), DER SubjectPublicKeyInfo, PKCS#1, raw 32-byte Ed25519 key
Revoked keys	An empty p= tag is detected and reported as a revoked key

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2. Requirements and Installation

Component	Purpose	Notes
Python 3	Runtime	Run with python / python3
cryptography	RSA / Ed25519 verification, key loading	Required. The script exits with an install hint if missing.
tkinter	Graphical interface	Bundled with most Windows/macOS Python installs; on Linux install python3-tk. Not needed with --no-gui.
Internet (HTTPS)	Mode 1 live DNS lookups	Only needed for Mode 1. Modes 2 and 3 work fully offline.

Get the tool and install the dependency:

```
git clone https://github.com/jjrboucher/DKIM-Validator.git
cd DKIM-Validator
pip install cryptography
```

Launching the tool

Launch method	Command	Result
Graphical (default)	python verify_dkim_bulk.py	Opens the configuration window
Command line	python verify_dkim_bulk.py --no-gui --src <folder> --out <folder> --mode <1|2|3>	Runs immediately (see Section 8)
Version	python verify_dkim_bulk.py --version	Prints the version and build date
Help	python verify_dkim_bulk.py -h	Lists command-line options

3. The Graphical Interface

The window is organised top-to-bottom in the order you configure a run.

3.1 Directory Workspace Configurations

Field	Default	Description
Source Folder	Current working directory	Folder containing the .eml files to verify. Use Browse... to pick a folder.
Output Folder	<current dir>/forensic_analysis_out	Where the CSVs, logs, keys and artifacts are written. Created if it does not exist.

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3.2 Directory Scan & Output Sub-Artifact Toggles

Option	Default	Effect
Scan Target Folder Recursively	On	Includes .eml files in all sub-folders. When off, only the top level of the Source Folder is scanned.
Extract & Save Canonicalized Body Artifacts (bodies/)	On	Saves the exact canonicalised body bytes that were hashed, one file per signature.
Extract & Save Cryptographic Header Blocks (signatures/)	On	Saves the reconstructed signed header block (the data the signature covers, with b= emptied), one file per signature.
Strict Expiry: Treat Expired Signatures (x= tag) as FAIL	Off	When on, a signature whose x= expiry has passed is reported FAIL (Expired). When off (forensic default) it can still PASS. Expiry is recorded for every signature either way (Section 6).

3.3 Public Key Mapping Strategy

Select one of three modes that determine where the public key comes from (Section 4).

Mode	Label in GUI	Key source
1	Automated Live DNS Resolution Query (w/ Local Cascading Fallbacks)	Live DNS, then local key files
2	Local 1:1 Sidecar File Matching Pattern	<EmailName>.key next to each .eml
3	Local N:1 Mapping Model	selector._domainkey.domain.key, then optional global key

Configure Global Manual Key Fallback (Optional Mode 3) opens a window where you paste key material (any format listed in Section 1) and click *Register Key Context*. This key is only used in Mode 3.

3.4 Content Threat Keyword Alerts Configuration

Ticking **Enable Advanced Body Content Keyword Flagging Alerts** reveals an editable box pre-filled with the default patterns (Section 7) and a **Load Custom Keywords File (.txt)** button that appends patterns from a text file. One regular expression per line.

3.5 Launch

LAUNCH TRIAGE PROCESSING LOOP validates the settings, closes the window and starts processing. Progress is shown in the console window. Before launching, the tool:

- Compiles every keyword line; a broken expression stops the launch and reports the offending line number.
- In Mode 3, warns if no global key is registered and no `*_domainkey*.key` files are found in the top level of the Source Folder, and asks whether to proceed.

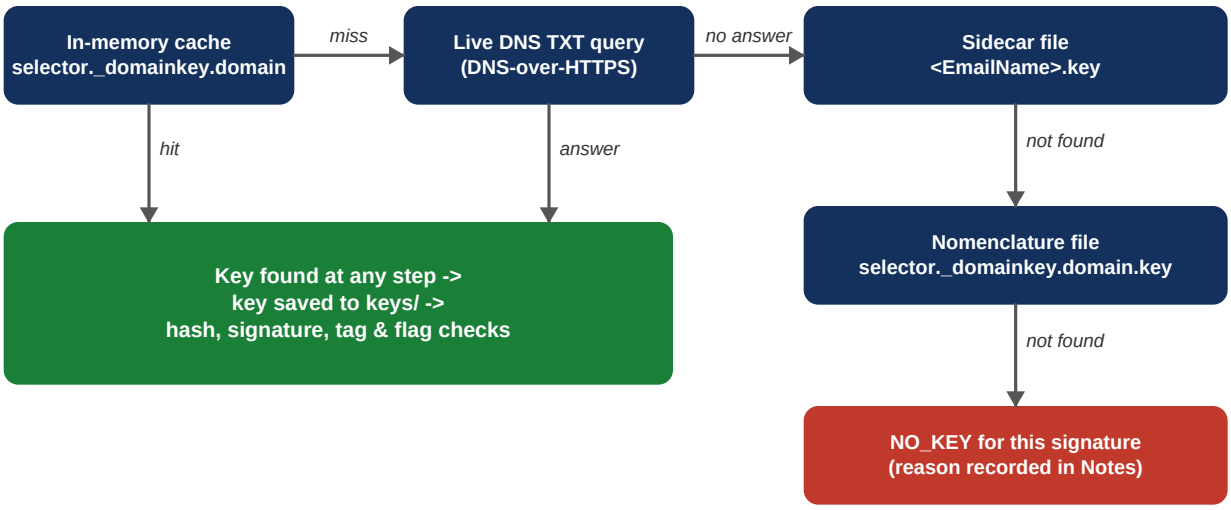
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4. Public Keys

DKIM keys are published in DNS at `<selector>._domainkey.<domain>`. Because domains rotate or retire keys, the key that signed an old email may no longer be in DNS. The three modes let you choose between the live record and keys you have preserved locally. When a message carries several signatures, the key is resolved separately for each signature's selector and domain.

4.1 Mode 1 - Live DNS with local fallbacks

Mode 1 - key resolution cascade (per DKIM signature)



- Lookups are sent over HTTPS to Google Public DNS (`https://dns.google/resolve`). Multi-string TXT records are joined; if several key records are published the first is used and noted.
- Successful and failed lookups are cached for the run, so each selector/domain pair is queried only once.
- Lookup failures (network error, NXDOMAIN, no TXT record, no p= tag) are recorded in Notes and the log before the local fallbacks are tried.
- Every key retrieved is saved to the output keys/ folder (Section 4.4).

> **Privacy and reproducibility**
> Mode 1 sends the selector and domain of each signature to a third-party resolver, and retrieves *today's* key, which may differ from the key in use when the email was sent.

4.2 Mode 2 - 1:1 sidecar files (offline)

Each email must have a key file with the same base name in the same folder. Only that file is used. If the email has several signatures, the sidecar key is tried against each; signatures made with a different key report FAIL (the email still PASSES if one signature verifies).

```
evidence/
  Invoice_2026-03-14.eml
  Invoice_2026-03-14.key     <- key for this email only
```

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4.3 Mode 3 - N:1 nomenclature files (offline)

One key file per selector/domain, shared by every email signed with that key, stored in the same folder as the emails. If no matching file exists, the Global Manual Key (if registered) is used. This mode handles multi-signature messages best, because each signature finds its own key file.

```
evidence/
  msg001.eml   (d=example.com; s=selector1)
  msg002.eml   (d=example.com; s=selector1)
  selector1._domainkey.example.com.key   <- used for both
```

> **Tip: build a preserved key library**
> Capture the DKIM TXT record at intake (e.g. nslookup -type=TXT selector1._domainkey.example.com) and save it as a nomenclature .key file. A Mode 1 run does this for you: its keys/ folder is already in nomenclature format.

Key_Source_Used values

Value	Meaning
CACHE_MEMORY	Mode 1 - key reused from an earlier lookup in this run
LIVE_DNS_QUERY	Mode 1 - key obtained from DNS
FALLBACK_SIDECAR_FILE	Mode 1 - DNS failed; <EmailName>.key used
FALLBACK_NOMENCLATURE_FILE	Mode 1 - DNS failed; selector._domainkey.domain.key used
SIDECAR_FILE	Mode 2 - <EmailName>.key used
N1_NOMENCLATURE_FILE	Mode 3 - selector._domainkey.domain.key used
GLOBAL_USER_SUPPLIED_KEY	Mode 3 - key pasted in the Global Manual Key window
UNKNOWN	No key located for the signature, or no signature present

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4.4 Preserving the keys used

In every mode, each distinct public key used during a run is saved to `<output>/keys/` as `<selector>._domainkey.<domain>.key`, containing the key record exactly as obtained. Every signature on every message is covered, so an email with three signatures contributes up to three keys. Each key is saved once per run, however many emails use it. If the same selector/domain resolves to different key material during a run (e.g. DNS differs from a local file), both are kept and the second is named `...domain__<12 hex of SHA-256>.key`.

keys_manifest.csv column	Content
Saved_Key_File, DNS_Name, DKIM_Selector, DKIM_Domain	Saved file and the selector/domain it belongs to
Key_Source_Used, Original_Key_File_Path	Where the key came from
First_Retrieved_UTC, Key_SHA256	When it was obtained, and its hash
Key_Type, Key_Bits	rsa / ed25519 and key size (e.g. 1024, 2048, 256)
Key_Testing_Mode, Key_Strict_Identity	YES / NO for the t=y and t=s flags
Key_Hash_Algs	h= list (e.g. sha256), or ANY
Key_Service_Types, Key_Notes, Key_Flags_Raw	s= (default *), n= text, raw t= value
Times_Used, Used_By	How often the key was used, and by which email#sigN

Both CSV ledgers have a `Saved_Key_File` column linking each signature to its preserved key, and each signature section of the log shows 'Key Preserved As', the key size and the key flags.

> **Re-running offline from preserved keys**
> Copy the files from keys/ into the folder holding the emails and run Mode 3. The results will match the original run regardless of later DNS changes, and Key_SHA256 will be identical. Revoked keys (empty p=) are saved too, as evidence of the state of DNS at the time.

4.5 Key record flags

A DKIM key record can carry flags that change how its signatures must be treated. The tool reads them from every key used, reports them in `Key_Flags` (both CSVs), the manifest and the log, and applies them as RFC 6376 requires:

Tag	Meaning	Tool behaviour
t=y	Domain is testing DKIM	Noted in Key_Flags and Notes. Result stands, but give it less weight: the domain has signalled its signatures should not be relied on
t=s	Strict identity	i= must use exactly the d= domain; a subdomain gives FAIL (Invalid i= tag)
h=	Permitted hash algorithms	If the signature's hash is not listed: FAIL (Key h= does not permit ...)
s=	Service types	Reported (default * = all)
n=	Free-text notes	Reported
Key size	From the key itself	Key_Bits; RSA keys under 1024 bits flagged in Notes (RFC 8301)

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> **Keys do not expire - signatures do**
> A key record has no expiry field. Expiry (x=) belongs to each signature and is recorded per signature (Section 6). A key used by many signatures may cover many different validity windows, so the manifest does not carry an expiry. Join the details CSV and the manifest on Saved_Key_File to see both. Domains retire a key by removing it from DNS or publishing an empty p= (revoked), which the tool reports.

5. Messages with Multiple DKIM Signatures

A message can carry several `DKIM-Signature` headers, and this is common. The tool examines **every** one of them; none are skipped.

Why messages have several signatures

Signer	Typical d=	Why
Author's own domain	example.org	The organisation signs its outgoing mail
Email service provider	mailer.example.net, sendgrid.net	Bulk/marketing platforms sign with their own domain as well
Mailing list or gateway	lists.example.com	Re-signs after modifying the message (Subject tag, footer)
Dual algorithms	example.org (twice)	Same domain signs with both RSA and Ed25519 keys

Each new signer adds its signature **above** the existing headers, so the top signature is the most recent and the bottom one is usually the first applied. The tool numbers them in file order: `Sig_Index 1` is the top signature.

How each signature is verified

Every signature is treated as a completely independent check, as RFC 6376 requires:

- its own public key, looked up from its own s= and d= (Section 4), and that key's flags;
- its own body canonicalisation and l= limit, compared with its own bh=;
- its own h= header list, rebuilt and checked against its own b= with its own algorithm (RSA or Ed25519);
- its own i=, t= and x= tags - so one signature can be expired while another is not.

A failed signature does not cancel a valid one. Under RFC 6376 a failing signature is treated as if it were not there, so the email's `Overall_Status` is **PASS if at least one signature passes**.

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What is recorded

Output	Multi-signature content
dkim_signature_details.csv	One row per signature with all fields, including expiry, and its own Signature_Status
Main CSV: Signature_Count / Signatures_Passed / Signatures_Expired	e.g. 3, 2 and 1
Main CSV: Reported_Signature	Which signature the per-signature columns show: the first that passed, otherwise the first that could be evaluated (e.g. '1 of 3')
Main CSV: All_Signatures	Every signature with its expiry and status, e.g. 1: d=lists.example.com ... x=2027-10-28 00:00:00Z [VALID] -> PASS | 2: d=mailer.example.net ... x=2026-10-30 10:20:03Z [EXPIRED] -> PASS | 3: ... x=none -> PASS
Log file	One section per signature, including the key record and flags
bodies/, signatures/	One artifact per signature, suffixed _sig1, _sig2, ...
keys/	The key used by each signature (Section 4.4)

Interpreting mixed results

Look at **which domain** passed, not just whether the email passed. A pass only for an intermediary's domain says less about the claimed sender than a pass for the domain in the From: header.

Pattern	Likely meaning
All signatures pass	Nothing covered by any signature has changed since the first signer
Only the top (list/gateway) signature passes	The intermediary changed the message (e.g. Subject tag, footer) and then re-signed. The content is vouched for by the intermediary only, from that point on
Author domain passes, intermediary fails	Usually an intermediary issue (key, configuration, or a header only it signed was changed); the author's signed content is intact
None pass	Modified after the last signature, wrong/rotated keys, or not a faithful copy

Key modes and multiple signatures

Mode	Behaviour
1	Each signature's key is looked up in DNS; failures fall back to local files per signature
2	The one sidecar key is tried against every signature; signatures made with a different key report FAIL. Not recommended for multi-signature messages
3	Each signature finds its own selector._domainkey.domain.key file. Recommended

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Multi-signature test pack

DKIM_MultiSig_TestPack contains five messages, each signed by three parties (bottom to top: example.org RSA, mailer.example.net RSA with a 7-day x=, lists.example.com Ed25519), and the three key files. Run it with Mode 3:

File	Expected	Passed	Demonstrates
01_multi_dkim_all_pass.eml	PASS	3 of 3	Three signers, RSA and Ed25519
02_multi_dkim_list_modified.eml	PASS	1 of 3	List tagged Subject and added footer, then re-signed
03_multi_dkim_body_tampered.eml	FAIL	0 of 3	Amount changed in the body
04_multi_dkim_LF_line_endings.eml	PASS	3 of 3	LF-only file normalised to CRLF
05_multi_dkim_from_header_altered.eml	FAIL	0 of 3	From: domain altered

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6. Signature Tags and Expiry

Signature tags

Tag	Check	Result if invalid
i=	Domain must equal d= or be a subdomain of it (RFC 6376 s3.5); exactly d= if the key has t=s. If absent, the default @d= is used	FAIL (Invalid i= tag)
t=	Converted to UTC. Flagged in Notes if in the future. The log shows its offset from the Date: header	Note only
x=	Must be later than t=. Compared with the verification time	FAIL (Invalid x= tag); expired: see below

How expiry is recorded

Expiry is recorded for **every** signature, whether or not Strict Expiry is on. Strict Expiry only decides whether an expired signature can still PASS.

Where	What
dkim_signature_details.csv	Sig_Timestamp_UTC, Sig_Expiry_UTC and Expiry_Status for each signature
Main CSV: All_Signatures	Each signature's expiry, e.g. x=2026-10-30 10:20:03Z [EXPIRED], or x=none
Main CSV: Signatures_Expired	Number of signatures in the email whose x= has passed
Main CSV: expiry columns	Expiry of the reported signature
Main CSV / log: Notes	'Signature expired (x=) at verification time; cryptographically valid' when an expired signature passes
Log	Signed (t=) and Expires (x=) [status] for each signature

Expiry_Status	Meaning
NO EXPIRY	The signature has no x= tag
VALID (not expired)	x= is later than the verification time
EXPIRED (at verification time)	x= has passed
INVALID (x= not later than t=)	x= is earlier than or equal to t= - the signature fails
INVALID (x= not an integer)	Malformed x= - the signature fails

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Effect of Strict Expiry

Situation	Strict Expiry off (default)	Strict Expiry on
Single signature, expired, otherwise valid	PASS	FAIL (Expired)
Several signatures, some expired, at least one valid and unexpired	PASS	PASS
Several signatures, every passing one expired	PASS	FAIL (Expired)
Expired signature that also fails the hash or signature check	FAIL	FAIL
Expired signature verifies, unexpired one fails	PASS	FAIL

Expired is not the same as altered
Many senders set x= only days after t=, so older evidence is often expired at verification time. An expired but cryptographically valid signature still shows the message has not been changed. RFC 6376 expects live mail receivers to reject expired signatures, which is what Strict Expiry reproduces; for forensic work keep it off and use the expiry columns to report the expiry.

Expiry test pack

DKIM_Expiry_TestPack contains eight messages and three key files: example.org (RSA 2048, h=sha256), mailer.example.net (RSA 1024, t=y, n= note) and lists.example.com (Ed25519). Run it in Mode 3, then again with Strict Expiry. Signatures are listed top to bottom.

File	Signatures	Default	Strict
01_single_rsa_expired	RSA, expired	PASS 1/1	FAIL (Expired)
02_single_ed25519_expired	Ed25519, expired	PASS 1/1	FAIL (Expired)
03_single_rsa_not_expired_control	RSA, valid to 2027	PASS 1/1	PASS 1/1
04_single_expired_and_tampered	RSA, expired; amount altered	FAIL	FAIL
05_multi_2sig_one_expired	ESP expired; author no x=	PASS 2/2	PASS 1/2
06_multi_2sig_both_expired	ESP expired; author expired	PASS 2/2	FAIL (Expired)
07_multi_3sig_mixed	List valid; ESP expired; author no x=	PASS 3/3	PASS 2/3
08_multi_only_expired_sig_verifies	ESP valid but Reply-To altered; author expired	PASS 1/2	FAIL

Sample 08 shows why Strict Expiry should stay off for forensic work: the only signature that still verifies is the expired one. The manifest also shows the ESP key as 1024-bit, testing mode, with its n= note.

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7. Keyword Alerts

Keyword alerts flag messages whose body matches investigator-defined regular expressions. They do not affect the DKIM result. Matching patterns are listed in the `Threat_Alerts` CSV column (or `CLEAN` if none match).

Default patterns

Theme	Pattern (case-insensitive)
Payment / banking	wire transfer | swift | routing number | bank account | invoice
Cryptocurrency	bitcoin | crypto | usdt | wallet phrase | private key
Credential phishing	password reset | verify your account | login expired | microsoft team
Legal / sensitive	subpoena | warrant | investigation | court notice | classified

Writing custom patterns

- One Python regular expression per line; blank lines are ignored.
- Prefix with `(?i)` for case-insensitive matching.
- Custom files are appended to whatever is already in the box - delete the defaults first if you do not want them.

```
(?i)(urgent payment|change of bank details)
(?i)\b(iban|bic)\b
(?i)gift ?cards?
```

> **Important: the scan runs on the raw body**
> Patterns are matched against the undecoded MIME body. Text inside base64 or quoted-printable encoded parts and attachments will usually not match. Treat CLEAN as 'no match in raw text', not as proof the content is benign.

8. Command-Line Reference

Argument	Required with --no-gui	Description
--no-gui	-	Skip the window and run from the command line
--src <folder>	Yes	Source folder containing .eml files
--out <folder>	Yes	Output folder for the CSVs, keys, logs and artifacts
--mode {1,2,3}	Yes	Public key mode (Section 4)
--strict-expiry	No	Treat signatures whose x= expiry has passed as FAIL (Expired)
--version	-	Print the version and build date
-h, --help	-	Show help

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Fixed settings in command-line mode

Setting	CLI behaviour
Recursive scan	Always on
Save bodies/ and signatures/	Always on
Key preservation (keys/, keys_manifest.csv)	Always on (all modes, GUI and CLI)
Keyword alerts	Off (Threat_Alerts = CLEAN)
Global manual key (Mode 3)	Not available - provide nomenclature .key files instead

Examples

```
# Live DNS with local fallbacks
python verify_dkim_bulk.py --no-gui --src C:\Case123\eml --out C:\Case123\dkim --mode 1

# Fully offline, one shared key file per selector/domain, strict expiry
python verify_dkim_bulk.py --no-gui --src ./eml --out ./dkim_out --mode 3 --strict-expiry
```

9. How Verification Works

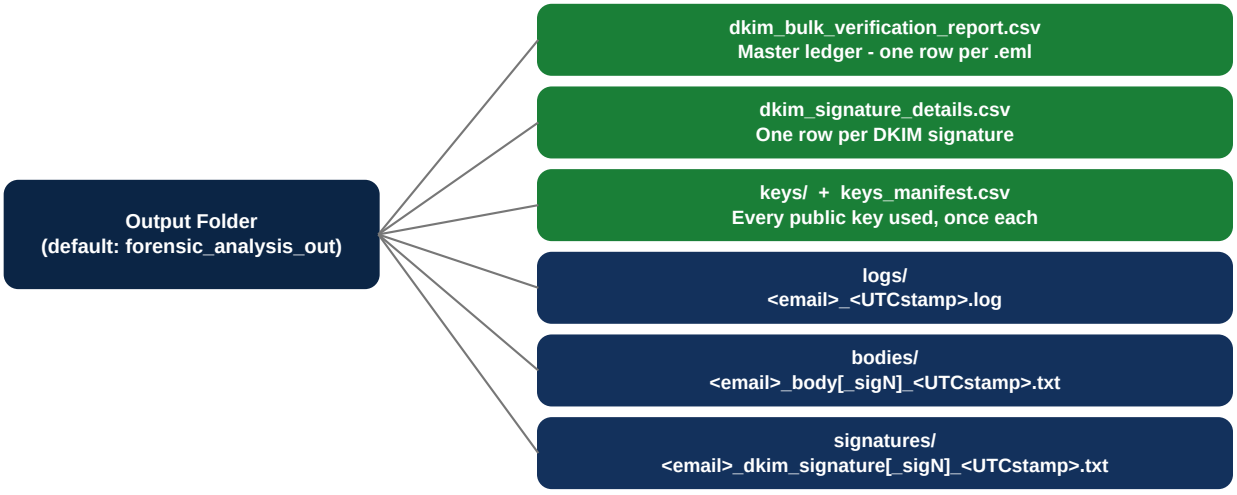
Step	Operation	Detail
1	Hash the file	MD5 and SHA-256 of the unmodified .eml bytes
2	Normalise line endings	Bare LF converted to CRLF on a working copy (reported in Line_Endings)
3	Split headers / body	At the first blank line
4	Find signatures	Every DKIM-Signature header, in order (top = most recently added)
5	Evaluate t= / x=	Converted to UTC; x= must be later than t=; expiry compared with verification time
6	Get public key	According to the selected mode, per signature; saved to keys/
7	Read key flags	t=y, t=s, h=, s=, n= and key size
8	Evaluate i=	Within d= (exactly d= if the key has t=s)
9	Canonicalise body	simple or relaxed per c=; truncated to l= if present
10	Body hash	SHA-256 (SHA-1 for rsa-sha1), base64; compared with bh=
11	Rebuild signed block	Headers in h= order (bottom-up for repeated names), then the DKIM-Signature with b= emptied
12	Verify signature	RSA: PKCS#1 v1.5 with SHA-256/SHA-1 over the block. Ed25519: over the SHA-256 digest of the block (RFC 8463)
13	Decide	Per signature PASS / FAIL (hash, signature, key h=, i=, x=, strict expiry); email PASSES if any signature passes
14	Record	CSVs, manifest, log, optional bodies/ and signatures/ artifacts

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> **Line-ending normalisation**
> SMTP transmits mail with CRLF line endings, and DKIM is calculated over CRLF data. Some export tools save .eml files with LF only. The tool converts bare LF to CRLF in memory before verifying. Hashes, the original file and Line_Endings reporting are unaffected; the saved bodies/ and signatures/ artifacts reflect the normalised data.

10. Output and Interpreting Results

Output folder layout



- Timestamps are UTC in the form YYYYMMDDTHHMMSSZ, so repeated runs do not overwrite logs or artifacts.
- Artifact names include `_sigN` when an email has more than one signature.
- All CSVs and the keys/ folder contents are overwritten on each run to the same Output Folder - use a new folder per run to keep history.
- A log is written for every signed email, including those with missing keys.

Main report columns (dkim_bulk_verification_report.csv)

Where an email has several signatures, the signature-specific columns describe the **reported signature**: the first one that passed or, if none passed, the first one that could be evaluated. All_Signatures and Signatures_Expired always cover every signature.

Column	Content
EML_Filename	File name of the message
MD5_Hash / SHA256_Hash	Hashes of the original .eml file
Line_Endings	CRLF, LF (normalised to CRLF) or MIXED (normalised to CRLF)
Signature_Count / Signatures_Passed / Signatures_Expired	Signatures found, passed, and with an expired x=
Reported_Signature	Which signature the following columns describe, e.g. '2 of 2'
DKIM_Domain / DKIM_Selector / DKIM_Algorithm	d=, s= and a= tags

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Column	Content
DKIM_Identity / Identity_Check	i= tag (or the @d= default) and whether it is valid
Sig_Timestamp_UTC / Sig_Expiry_UTC / Expiry_Status	t= and x= in UTC and the expiry state (Section 6)
Key_Source_Used / Key_File_Path / Key_SHA256	Where the key came from, local file used, SHA-256 of the key text
Saved_Key_File	Preserved copy of the key in keys/ (Section 4.4)
Key_Bits / Key_Flags	Key size, and key record flags (t=y, t=s, h=, s=, n=) or NONE (Section 4.5)
Body_Hash_Result	MATCH / MISMATCH / ERROR / N/A
Signature_Result	VERIFIED / FAILED / ERROR / N/A
Overall_Status	See status table below
All_Signatures	Every signature with its expiry and status
Threat_Alerts	Matching keyword patterns, or CLEAN
Notes	DNS lookup results, expiry, testing-mode keys, l= coverage, deprecated algorithm and weak key warnings

Signature details (dkim_signature_details.csv)

One row per signature with the same per-signature fields, plus `Sig_Index`, `Canonicalization` and `Signature_Status`. Use it when an email has several signatures, or to filter by domain or expiry.

Status values

Overall_Status (per email)	Meaning
PASS	At least one signature passed every check
FAIL	No signature passed
FAIL (Expired)	Strict Expiry on; every evaluated signature would pass but has expired
UNVERIFIED (No Signature)	Message has no DKIM-Signature header
UNVERIFIED (Missing Public Key)	No key found for any signature
RUNTIME_ERROR (...)	Processing error, e.g. malformed file, revoked/invalid key; message included

Signature_Status (per signature)	Meaning
PASS	Body hash MATCH, signature VERIFIED, tags and key flags satisfied (and not expired in strict mode)
FAIL	Body hash mismatch and/or signature failed
FAIL (Key h= does not permit ...)	Cryptographically valid, but the key's h= list excludes the signature's hash
FAIL (Invalid i= tag)	Cryptographically valid, but i= is outside d= (or not exactly d= with key t=s)
FAIL (Invalid x= tag)	Cryptographically valid, but x= is not later than t= (or not a number)
FAIL (Expired)	Strict Expiry only: valid but x= has passed

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Signature_Status (per signature)	Meaning
NO_KEY	No public key found for this signature
ERROR (...)	The signature could not be evaluated; reason included

Reading a result

Body hash	Signature	Likely interpretation
MATCH	VERIFIED	Signed headers and body unchanged since signing
MISMATCH	VERIFIED	Signed headers intact but body changed - e.g. edited content, or altered in transit by a gateway/disclaimer/footer
MATCH	FAILED	Body intact but a signed header changed, wrong/rotated key, or export altered headers
MISMATCH	FAILED	Message modified, wrong key, or file not a faithful copy

A FAIL is not automatically proof of tampering
Legitimate causes include mailing-list or security-gateway modification, key rotation (Mode 1 retrieves today's key) and exports that re-encode the message. Corroborate with headers, server logs and other copies of the message.

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11. Known Limitations (v1.7.2)

Area	Observation	Impact / workaround
Mode 2 with several signatures	A single sidecar key is tried against every signature.	Signatures from other domains report FAIL; use Mode 3 for multi-signature messages.
Keyword scan	Runs on the raw, undecoded body.	Encoded content will not match (Section 7).
Mode 3 pre-check	The GUI warning looks only at the top level of the Source Folder.	Nomenclature files must sit beside the emails they apply to.
Policy checks	DMARC/SPF, From-domain alignment and whether h= includes From are not assessed.	Compare DKIM_Domain with the From header manually when impersonation is in question.
Mode 1	Retrieves the current key from a third-party resolver.	Prefer Mode 3 with keys preserved at intake for repeatable results.

Good practice

- Work on copies of evidence; record the SHA-256 values in the CSV against your intake hashes.
- Prefer Mode 3 with key records preserved at intake for repeatable results.
- Keep the CSVs, keys, logs and artifacts from each run together in a dedicated output folder.

12. Version History

Version	Date	Changes
1.7.2	2026-11-05	All_Signatures shows each signature's x= expiry and state; new Signatures_Expired column. Key record evaluation: Key_Bits and Key_Flags columns; manifest gains key size, testing mode, strict identity, hash algorithms, service types, notes and raw flags. Key flags enforced: t=s (exact i= domain), h= (permitted hashes); t=y noted; RSA keys under 1024 bits flagged. Guide: Section 4.5 Key record flags, Section 6 Signature Tags and Expiry, expiry test pack.
1.7.1	2026-11-05	Every public key used preserved in keys/ with keys_manifest.csv; Saved_Key_File column; caches reset per run. Guide: section on messages with multiple DKIM signatures; multi-signature test pack.
1.7.0	2026-11-05	Fixed Mode 1 DNS-over-HTTPS URL; DNS errors reported; failed lookups cached; multi-string TXT joined. All DKIM-Signature headers verified; dkim_signature_details.csv. LF-only and mixed line endings normalised. Ed25519 verified per RFC 8463. t=, x= and i= evaluated; Strict Expiry option. Notes column, --version, Apache 2.0 header.
1.6.2	2026-11-05	Initial documented release

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13. License, Warranty and Repository

Repository

The script, this guide and the README are published at https://github.com/jjrboucher/DKIM-Validator. Check the repository for the latest version, release notes and issue reporting.

Licensing

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Suggested attribution for this guide

DKIM Bulk Signature Verifier - User Guide (v1.7.2) by Jacques Boucher, licensed under CC BY 4.0. https://github.com/jjrboucher/DKIM-Validator

Tool	DKIM Bulk Signature Verifier
Version	1.7.2
Build date	2026-11-05
Script	verify_dkim_bulk.py
Author	Jacques Boucher (jjrboucher@gmail.com)
Repository	https://github.com/jjrboucher/DKIM-Validator
Licenses	Software: Apache-2.0 | Documentation: CC BY 4.0